# Coldstone Web data pipeline

AWS cost summary: what each service costs and why

| about $1 | about $15 | about $16 |
|---|---|---|
| per month for everything except Redshift (Glue, S3, Lambda, alerts, logs) | per month for the Redshift loads, on the eatngo-analytics workgroup at 8 RPU (section 3) | per month in total, before report and dashboard queries |

The pipeline runs once a day for Nigeria and Kenya: it pulls the previous day's data from the Coldstone API, stores it in S3, loads it into Redshift and posts a Teams summary. Almost every part is billed only while it runs, and the volumes are small (a few hundred orders a day), so the running cost is low. **Redshift is the only noticeable cost**, and it is shared with the other pipelines on the same workgroup.

**These are estimates** from AWS list prices for us-east-1 (N. Virginia) and the run times observed while building the pipeline. Prices change; the AWS pricing pages are authoritative. After the first full month, **Billing and Cost Management, Bills** shows the actual charge per service.

## 1. Cost by service

Monthly estimates for both countries together, at today's volumes.

| Service | What it does in this pipeline | How AWS charges | Est. per month |
|---|---|---|---|
| Redshift Serverless | Warehouse: loads each day's data and serves queries (database coldstone_web) | Per second of compute while queries run (RPU-hours), 60-second minimum per query; plus storage per GB-month | about $15 for the loads, at 8 RPU (section 3) |
| AWS Glue | Runs the 4 jobs: extract NG / KY (API to S3) and load NG / KY (S3 to Redshift) | Per second while a job runs, 1-minute minimum, at 1/16 DPU (the smallest size) | about $0.40 |
| Amazon S3 | Stores the raw API responses and the processed Parquet files | Per GB stored per month, plus a small fee per request | under $0.10 (grows by roughly 0.5 GB a month) |
| CloudWatch | Job and Lambda logs; the 'load requests stuck' alarm | Per GB of logs, per alarm | under $0.30 |
| AWS Lambda | Posts Teams cards; starts loads from the queue | Per request and compute time; 1 million requests a month free | $0 (free tier) |
| Amazon SNS | Carries pipeline results and alerts | Per million messages; first million free | $0 (free tier) |
| Amazon SQS | Load queue and dead-letter queues | Per million requests; first million free | $0 (free tier) |
| EventBridge | Forwards Glue failures to the alerts topic | AWS service events on the default bus are free | $0 |
| SSM Parameter Store | Holds the API keys and Teams webhook | Standard parameters are free | $0 |
| **Total (excluding report queries)** | | | **about $16** |

## 2. AWS Glue in detail

Glue charges **DPU-hours**: the job's size (in DPUs) multiplied by how long it runs, at about **$0.44 per DPU-hour**. All four jobs use **1/16 DPU**, the smallest Python shell size, so a full hour costs under 3 cents. Nothing is charged while jobs are not running.

| Job | Typical run | Cost per run | Per month (30 runs) |
|---|---|---|---|
| coldstone-extract-ng | about 15 min | $0.007 | $0.21 |
| coldstone-extract-ky | about 4 min | $0.002 | $0.06 |
| coldstone-load-ng | about 5 min | $0.002 | $0.07 |
| coldstone-load-ky | about 4 min | $0.002 | $0.06 |
| **Total** | | **about $0.013 a day** | **about $0.40** |

Formula: 0.0625 DPU x run time in hours x $0.44. Retries, reruns and the occasional full-history run add only cents. Start-up time counts, including about a minute on the load jobs to install a newer AWS library.

## 3. Redshift Serverless in detail

Redshift Serverless bills **compute only while queries are running**, per second, with a 60-second minimum each time it wakes up. Compute is measured in RPU-hours (Redshift Processing Units) at about **$0.375 per RPU-hour**. When nothing runs, compute costs nothing. Storage is billed separately at about $0.024 per GB-month.

The eatngo-analytics workgroup runs at a **base capacity of 8 RPU**, the smallest size Redshift Serverless offers. The two loads keep it busy for about 10 minutes a day in total, which comes to about **$15 a month**. The table shows how the same work would cost at larger sizes, for reference:

| Workgroup base capacity | Cost per hour while running | Daily loads (about 10 min) | Per month |
|---|---|---|---|
| **8 RPU (current setting)** | **$3.00** | **about $0.50** | **about $15** |
| 16 RPU | $6.00 | about $1.00 | about $30 |
| 32 RPU | $12.00 | about $2.00 | about $60 |
| 128 RPU (AWS default) | $48.00 | about $8.00 | about $240 |

**Already at the minimum:** 8 RPU is the smallest base capacity, so there is no saving to make on workgroup size. Because the workgroup is shared with the pulse pipelines, its total bill also includes their queries; the figure here is the share caused by the Coldstone loads. If Redshift is ever busy, it scales above 8 RPU for that query only and back down afterwards.

**Storage:** the Coldstone tables are well under 1 GB, so storage is a few cents a month. **Report and dashboard queries** are billed the same way as the loads (compute while they run), so their cost depends on how often reports refresh; this is the main thing that can grow.

## 4. What keeps the cost low

| Choice | Effect |
|---|---|
| Glue Python shell at 1/16 DPU, not Spark | Spark jobs start at 2 DPUs (often 10), which would cost 30-150 times more for the same work. |
| Pay-per-run services throughout | Glue, Lambda, SNS, SQS and Redshift compute cost nothing between runs. |
| One daily run (D-1), loading only what changed | Each load touches a day of data, not the whole history. |
| Job timeouts (45-60 min) and one run at a time | A stuck job can never run up a large bill. |

| Choice | Effect |
|---|---|
| Retries and a queue instead of bigger capacity | A busy Redshift is handled by waiting and retrying, not by paying for more compute. |
| S3 lifecycle rule (old versions deleted after 30 days) | Reruns don't keep paying for superseded files. |
| Parquet format in S3 | Files are about 10 times smaller than CSV and load faster. |

## 5. Recommended cost controls

| Action | Where | Why |
|---|---|---|
| Keep the Redshift base capacity at 8 RPU | Redshift, Workgroup configuration, eatngo-analytics, Limits | Already the minimum; raising it would multiply the Redshift cost. |
| Set a Redshift usage limit (e.g. daily RPU-hours, action: alert) | Same Limits tab | Warns before an unexpected query runs up cost. |
| Create an AWS Budget for the account or tagged resources | Billing and Cost Management, Budgets | Email alert if monthly spend passes a set amount. |
| Review the bill after the first full month | Billing and Cost Management, Bills | Replaces these estimates with actual figures. |
| Keep report refreshes reasonable (e.g. hourly, not every minute) | BI tool settings | Report queries are the cost that can grow. |

Assumptions: us-east-1 list prices; both countries running daily; run times as observed in October 2026 (Nigeria roughly 100-150 orders a day, Kenya a few a day); no report or dashboard queries included. Costs scale roughly with run time, so they grow slowly as order volumes and branch counts grow.